

Instructor: Date:

Learner: Regional Branch:

Program: Center:

CONTENT

- Let's Start! • Let's Explore! • Let's Practice! • Extension Activity • Target Words • Learning Points



Let's Start!

All About Structured Data Types



Learning Outcome:

Describe common activities at work to a visitor using short sentences in familiar spoken and written situations.
Understand information about data and its structure in simple spoken and written situations.

Can Do Statements:

- I can describe structured data types.
- I can give detailed information related to structured data types.
- I can talk about key features related to structured data types.

Ask your partner:

- What do you know about modern programming languages?
- Which data types do you know?





Let's Explore!

Daniel and Carol are studying for their test.



Instructions: Read the text below.

- Carol:** Hi, are you ready for today's exam?
- Daniel:** I don't feel ready, but I must pass it!
- Carol:** I could be the one who helps you. We have to study a lot! Let's review a key term first, what's a data type?
- Daniel:** Well, it's a classification that programmers do according to the use they want to give to the data. A type can define the meaning, the usage, and the processes for storing the different values within a piece of data.
- Carol:** See? You're clever, you're a bookworm! Now, what's a structured data type?
- Daniel:** I've been studying eagerly. Structured data types are common in modern computer languages. You know that the data types that we've studied so far can take just one value, right? Unlike those, structured data types can take more than a piece of data.
- Carol:** Now I remember. The array is a basic structured data type.
- Daniel:** That one can hold a collection of variables or values.
- Carol:** We can say then that structured data types help define complex data structures. Do you have another example?
- Daniel:** Remember the programming language called Pascal? There you can find a structured data type known as record.
- Carol:** That one is used for collecting different data types that define a single value. For example, if you want to define a pillow, it'll give information on the size, the fabric, and the color.
- Daniel:** That one is a good example. The teacher told us that when using Pascal, we needed to write the code carefully because the ones who are new in this programming language tend to forget the end keyword. Can you give me more examples of structured data types?
- Carol:** The Boolean data type and the Struct in C programming, which is a composite data type.
- Daniel:** Good! I think we're ready! I would feel like a million dollars if I pass the exam!



Let's Practice!

Daniel is giving valuable information on structured data types. Is he using an adjective or an adverb?

Instructions: Select the correct category in the table. There is only one answer per category.

	Adverbs	Adjectives
Structured data types are useful because they can take more than one value.		
When you use a record in Pascal, you have to write the code carefully .		
Structured data types are used with complex data structures.		
The struct in C programming and the Boolean data type are good examples of structured data types.		

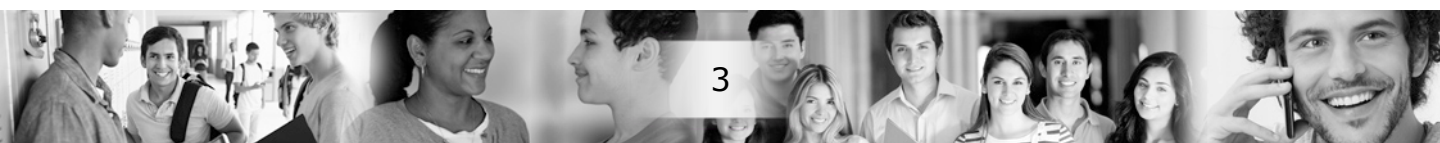


Let's Practice!

Let's go over some important features of structured data types

Instructions: Select all the correct answers.

- Data types ____ take just one value.
 - cannot
 - should not
 - must
 - can
- A record in Pascal is a structured data type. You ____ include the end keyword when writing the code.
 - must
 - can
 - cannot
 - do not
- Structured data types ____ take more than one value.
 - cannot
 - can
 - are
- In C programming, you ____ find an example of a structured data type, which is called struct.
 - must
 - can
 - are not
 - should





Extension Activity

Examples of structured data types.

Instructions: Complete the following steps. Fill in the Self-Assessment.

- Look for more information of examples related to structured data types.
- Write a paragraph in which you describe each example and the reason why they are structured data types.
- Be careful with spelling and grammar. Make room for the grammatical topics of the lesson (adjectives vs. adverbs, noun clauses and pronouns one and ones).
- Share your paragraph with a partner.

Self-Assessment

Yes No

- I look for more examples of structured data types.
- I write a paragraph including and describing those examples.
- I try to include the grammar topics of the lesson and check spelling.
- I read my paragraph to a partner.

☐	☐
☐	☐
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☐	☐



Target Words

- eagerly** (adverb): it's used for describing when something's done with enthusiasm.
e.g. I've been studying eagerly.
- programmer** (noun): a person who writes code for computer programs or devices
e.g. It's a classification that programmers do according to the use that they want to give to the data.
- feel like a million dollars** (expression): to feel extremely happy or good
e.g. I would feel like a million dollars if I pass the exam.
- pass** (verb): to have a good result in a test or exam
e.g. I must pass it. I don't feel ready.
- piece of data** (noun): a small part of information
e.g. A type can define the meaning, the usage and the processes for storing the different values within a piece of data.
- bookworm** (noun): someone who studies or reads a lot
e.g. You're a bookworm!



Learning Points

Pronouns One and Ones

We use **one** for singular and **ones** for plural. Using these pronouns, we avoid repeating a noun that was mentioned before.

We use **one** if we talk about a singular noun, for example:

A: Could you give me **the book**?
B: Yes, the **one** with the hardcover?
A: Did you see **the teacher**?
B: Which **one**? The **tall one**?

We use **ones** if we want to avoid repeating a plural noun, take a look at the following examples:

A: Did you see **my notes**?
B: Which **ones**?
A: Look at **those shoes**!
B: Which ones? The **black ones**?

Considering the fact that we already know what we are talking about, we use **one** or **ones** in order to avoid unnecessary repetitions.

Examples

- *I could be the one who helps you.*
- *That one can hold a collection of variables or values.*
- *That one is used for collecting different data types that define a single thing.*
- *(...) the ones who are new in this programming language tend to forget the end keyword.*

Noun Clauses

Think of a **noun clause** as a group of words that work together as a single noun, which can be a subject, object or complement.

A **noun clause** is called **dependent or subordinate clause**. It contains a subject and a verb but does not express a whole idea. As a result it's only a fragment, it can't stand by itself.





Learning Points

Noun clauses begin with words such as:

How, that, what, whatever, when, where, whether, which, who and why.

For example:

- *Once, we understand this topic, we can do **whatever we want**.*
- *We have to know **how this works**.*
- *Do you know **what the next topic will be**?*
- *I'm reading **that structured data types are easy**!*
- *Our teacher told me **that today's lesson is cancelled**.*
- *I like **what I read**.*

Examples

- *You know that the data types that we've studied so far can take just one value, right?*
- *That one is used for collecting different data types that define a single thing.*

Grammar Concerns: Adjectives vs. Adverbs

When learning a language, it's important to know the difference between adjectives and adverbs. They are both used to describe, but you have to know when to use each.

Think of **adjectives** as the words in charge of giving a characteristic to a noun or pronoun, for example:

- *An **interesting** book*
- *A **long** and **boring** lesson*
- *She is a **hardworking** student.*

Interesting, long, boring and **hardworking** are **adjectives** used to describe nouns (a book, a lesson or a woman).

Now, **adverbs** are used to describe actions or verbs. Let's consider the following sentences:

- *She works **patiently**.*
- *You need to finish your assignment **quickly**.*





Learning Points

Notice that ***patiently*** does not refer to the woman performing the action (*to work*), but to the action itself. The same thing happens in the second sentence, because the adverb ***quickly*** refers to the verb to finish.

Take into account the following list of some basic adjectives and adverbs that have the same meaning:

Adjectives	Adverbs
Good	Well
Bad	Badly*
Perfect	Perfectly*
Safe	Safely*
Hard**	Hard
Loud	Loudly*
Immediate	Immediately*

*Notice you can write an adverb using an adjective as a root and adding the suffix ***-ly*** at the end.

There are some words like *hard***, ***late***, ***clear*** or ***fast*** that are both adjectives and adverbs. Be careful with these exceptions.

Examples

- *Are you ready for today's test?*
- *See? You're clever.*
- *I've been studying eagerly.*
- *Structured data types are common in modern computer languages.*
- *We can say then that structured data types help define complex data structures.*
- *We needed to write the code carefully.*